

OSN2526: Assignment 2: Page tables

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10 april 2026

1 Inleiding

This report describes an analysis of different virtual memory implementations. In Section 2 a comparative analysis is performed between a simple single level MMU and a RISC-V Sv48 MMU. Both MMU are simulated within the provided framework given for the assignment.

2 MMU Comparison

This section performs a comparative analysis of a simple single level MMU provided for the assignment and a RISC-V Sv48 MMU implementation.

Table 1 shows the results for a single process.

Tabel 1: Single process results of RISC-V Sv48 and Simple MMU Implementations for trace: gcc.txt.gz.

Metric	RISC-V (Sv48)	Simple (Single-Level)
Page Size	4 KB	64 MB
Page Table Allocated	36 KiB	64 MiB
Handled Page Faults	6	4
Max. Allocated Physical Pages	15	5

Tabel 2: Multiprocess results of RISC-V Sv48 vs. Simple MMU (gcc.txt.gz and ls.txt.gz)

Metric	RISC-V (Sv48)	Simple (Single-Level)
Page Size	4 KB	64 MB
Page Table Allocated	68 KiB	128 MiB
Handled Page Faults	11	7
Max. Allocated Physical Pages	28	8

3 Conclusie